

AZURE DATA ENGINEER – SCENARIO INTERVIEW QUESTIONS

150 Real-Project Scenarios | 5–10 Years Experience | Interview Preparation

- Q1.** How would you design an Azure data platform that ingests data from SQL Server, APIs, SaaS applications, and files into a centralized lake for analytics?
- Q2.** Your organization receives several terabytes of daily data from multiple business systems. How would you design a scalable ingestion architecture on Azure?
- Q3.** A critical data pipeline must support both batch and near-real-time workloads. How would you design the overall architecture and decide which services to use?
- Q4.** How would you design a medallion-style data architecture using Azure Data Lake Storage Gen2 for raw, cleansed, and curated data?
- Q5.** A business team wants a single trusted source for analytics, but different systems contain conflicting customer data. How would you design the data integration and reconciliation process?
- Q6.** How would you design an Azure data platform for multiple business domains while keeping ownership, security, and governance decentralized?
- Q7.** A data platform is growing rapidly and storage and compute costs are increasing. How would you identify the main cost drivers and redesign the architecture?
- Q8.** How would you design a highly available Azure data platform with appropriate backup, disaster recovery, and regional failover capabilities?
- Q9.** Your company is migrating an on-premises data warehouse to Azure. How would you assess the existing platform and plan the migration with minimal business disruption?
- Q10.** How would you design a data platform that can handle schema changes from source systems without repeatedly breaking downstream pipelines?

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- Q11.** An Azure Data Factory pipeline is taking much longer than expected. How would you troubleshoot where the bottleneck is occurring?
- Q12.** A pipeline loads millions of rows from SQL Server into ADLS Gen2 every night. How would you improve throughput and reduce the load window?
- Q13.** How would you implement incremental loading in Azure Data Factory when the source system contains a reliable last-modified timestamp?
- Q14.** The source database does not provide a reliable watermark column. How would you design incremental extraction without repeatedly copying the entire table?
- Q15.** How would you design a metadata-driven Azure Data Factory framework that can onboard hundreds of source tables with minimal manual configuration?
- Q16.** A single ADF pipeline contains hundreds of activities and has become difficult to maintain. How would you refactor it?
- Q17.** How would you design parameterized and reusable ADF pipelines for different environments such as development, test, and production?
- Q18.** A pipeline succeeds for most source tables but fails intermittently for a few large tables. How would you investigate and stabilize it?
- Q19.** How would you design retry, timeout, dependency, and failure-handling logic for a business-critical ADF pipeline?
- Q20.** How would you prevent duplicate records when an ADF pipeline is accidentally triggered twice for the same business date?

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Q21. How would you design a secure connection from Azure Data Factory to an on-premises SQL Server through a corporate network?

Q22. Your self-hosted integration runtime frequently goes offline. How would you troubleshoot the issue and design for higher reliability?

Q23. When would you choose Azure Integration Runtime versus Self-hosted Integration Runtime for an ADF workload?

Q24. How would you securely connect ADF to Azure SQL Database, ADLS Gen2, Key Vault, and other Azure resources without storing passwords in pipeline definitions?

Q25. An ADF pipeline needs to access hundreds of files in ADLS, but processing every file is too expensive. How would you design efficient file discovery and processing?

Q26. How would you process only newly arrived files from a landing zone while ensuring that files are not processed twice?

Q27. A source system drops files with inconsistent names and formats. How would you build a robust ADF ingestion process around them?

Q28. How would you design a pipeline to move data from multiple REST APIs when each API has pagination, rate limits, and different authentication mechanisms?

Q29. An API occasionally returns HTTP 429 responses. How would you design ADF or surrounding orchestration to handle throttling gracefully?

Q30. How would you securely store and rotate API credentials used by Azure Data Factory?

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Q31. How would you design an ADLS Gen2 folder and naming structure for thousands of datasets owned by multiple teams?

Q32. Your ADLS account contains billions of small files and downstream Spark jobs are becoming slow. How would you address the small-file problem?

Q33. How would you decide between CSV, JSON, Parquet, and Delta formats for different layers of an Azure data lake?

Q34. How would you design partitioning for a large fact dataset stored in ADLS so that queries remain efficient as data grows?

Q35. A query scans terabytes of data even though users filter by date. What would you investigate in the lake storage and table design?

Q36. How would you implement lifecycle management so that old data moves to lower-cost storage while remaining compliant with retention requirements?

Q37. How would you protect sensitive data stored in ADLS from unauthorized access while allowing multiple analytics teams to work with it?

Q38. How would you design access control in ADLS using Azure RBAC and POSIX-style ACLs for different teams and datasets?

Q39. A data lake has accumulated duplicate, obsolete, and temporary files. How would you design a safe cleanup strategy?

Q40. How would you design a disaster-recovery strategy for critical data stored in ADLS Gen2?

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Q41. How would you design a dimensional data warehouse on Azure for sales reporting with fact and dimension tables?

Q42. A dimension receives corrections to customer attributes after historical transactions have already been loaded. How would you implement an appropriate slowly changing dimension strategy?

Q43. When would you choose a star schema versus a normalized model for an analytics workload?

Q44. A large fact table is causing slow BI queries. How would you investigate and optimize the warehouse model?

Q45. How would you design surrogate-key generation for dimensions in a distributed data platform?

Q46. How would you handle late-arriving dimension records when fact data arrives before the corresponding dimension?

Q47. A business requires historical reporting exactly as the data looked at the time of each transaction. How would you design the data model?

Q48. How would you design a warehouse load so that a failed batch can be safely restarted without corrupting previously loaded data?

Q49. How would you validate source-to-target row counts, totals, and business rules during a warehouse load?

Q50. How would you design a reconciliation framework that identifies missing, duplicate, and mismatched records between source and warehouse?

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Q51. How would you design an Azure Synapse Analytics dedicated SQL pool for a large enterprise data warehouse?

Q52. A Synapse query has suddenly become much slower after the data volume increased. How would you diagnose and tune it?

Q53. How would you choose an appropriate distribution strategy for a large fact table in Synapse?

Q54. A Synapse table has severe data skew. How would you identify the cause and redesign the table?

Q55. When would you choose hash, round-robin, or replicated distribution in Synapse?

Q56. How would you design table partitioning in Synapse for a very large time-series fact table?

Q57. How would you optimize a Synapse workload with frequent joins between large fact and dimension tables?

Q58. How would you load large volumes into Synapse efficiently while minimizing impact on concurrent reporting users?

Q59. How would you design workload management so that ETL jobs do not consume all resources needed by BI users?

Q60. How would you approach performance troubleshooting when a Synapse query plan shows unexpected data movement?

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- Q61.** How would you design a serverless SQL solution in Synapse for querying files in ADLS without loading all data into a dedicated warehouse?
- Q62.** A business wants ad-hoc SQL queries over Parquet files in the data lake. How would you design the solution?
- Q63.** A serverless Synapse query is expensive because it scans too much data. How would you reduce the amount of data processed?
- Q64.** How would you expose curated lake data to analysts while preventing them from directly accessing sensitive raw files?
- Q65.** When would you choose Synapse serverless SQL over dedicated SQL pools for an analytics workload?
- Q66.** How would you design external tables or views over lake data for consistent analyst access?
- Q67.** A serverless query fails because files have inconsistent schemas. How would you investigate and make the data consumable?
- Q68.** How would you optimize Parquet data layout for serverless SQL workloads?
- Q69.** How would you control and monitor serverless SQL costs across multiple teams?
- Q70.** How would you design a hybrid architecture using lake storage, serverless SQL, and dedicated warehouse compute?

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Q71. How would you design an Azure Databricks solution for large-scale data transformation using Apache Spark?

Q72. A Databricks Spark job is running for hours instead of minutes. How would you systematically troubleshoot and optimize it?

Q73. How would you identify and resolve data skew in a Spark transformation?

Q74. A Spark job is failing with out-of-memory errors. What would you investigate before simply increasing cluster size?

Q75. How would you choose between job clusters, all-purpose clusters, and serverless compute for different workloads?

Q76. How would you design Databricks notebooks and jobs so that business logic is reusable across many pipelines?

Q77. How would you implement incremental processing in Databricks for data arriving continuously in a data lake?

Q78. How would you optimize joins between a very large dataset and a relatively small reference dataset in Spark?

Q79. A Databricks job creates thousands of tiny output files. How would you redesign the write process?

Q80. How would you monitor Databricks jobs for failures, long runtimes, excessive compute usage, and data-quality issues?

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Q81. How would you design Delta Lake tables for reliable ACID-based data processing in an Azure data platform?

Q82. A Delta table contains duplicate records because an upstream process was retried. How would you make the ingestion idempotent?

Q83. How would you implement MERGE logic for upserting changed records into a Delta table?

Q84. How would you handle schema evolution in Delta tables when new columns are introduced by the source system?

Q85. A Delta table has accumulated many obsolete files after repeated updates. How would you manage storage and optimize the table safely?

Q86. How would you design a bronze-silver-gold Delta architecture for an enterprise analytics platform?

Q87. How would you recover a Delta table after an incorrect transformation overwrote valid data?

Q88. How would you design data-quality checks before promoting data from a raw layer to a curated layer?

Q89. How would you handle deletes from a source system when the source provides only changed records and not a full snapshot?

Q90. How would you design retention and recovery policies for Delta tables while balancing storage cost and audit requirements?

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Q91. How would you design a real-time ingestion architecture using Azure Event Hubs for high-volume application events?

Q92. An Event Hubs consumer is falling behind during traffic spikes. How would you identify the bottleneck and scale the processing layer?

Q93. How would you design partitioning and consumer groups for an Event Hubs workload with multiple downstream consumers?

Q94. How would you ensure that real-time event processing can tolerate duplicate messages and occasional out-of-order events?

Q95. When would you choose Event Hubs versus Service Bus for a data engineering workload?

Q96. How would you design a streaming pipeline that writes Event Hubs data into ADLS for both real-time and historical analytics?

Q97. A streaming pipeline loses connectivity temporarily. How would you design checkpointing and recovery so that events are not unnecessarily lost or duplicated?

Q98. How would you monitor streaming latency, throughput, failed events, and consumer lag in production?

Q99. How would you design dead-letter or quarantine handling for malformed streaming events?

Q100. How would you build a near-real-time dashboard while also retaining the complete event history for later analytics?

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Q101. How would you design an Azure Stream Analytics job to transform and aggregate streaming telemetry before sending it to downstream systems?

Q102. A Stream Analytics job is producing delayed results during a traffic spike. How would you troubleshoot and tune it?

Q103. How would you handle late-arriving events and out-of-order events in a streaming analytics workload?

Q104. How would you design windowed aggregations for hourly and daily business metrics from a continuous event stream?

Q105. A streaming query produces incorrect counts because duplicate events are received. How would you design deduplication?

Q106. How would you safely deploy changes to a production streaming job without causing unacceptable data loss?

Q107. When would you use Databricks Structured Streaming instead of Azure Stream Analytics?

Q108. How would you design a streaming pipeline that enriches events with customer or product reference data?

Q109. How would you detect malformed events in a streaming pipeline and prevent them from blocking valid events?

Q110. How would you validate real-time metrics against batch-calculated historical metrics to detect discrepancies?

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Q111. How would you design data quality checks for completeness, uniqueness, validity, consistency, and freshness across an Azure data platform?

Q112. A pipeline reports success, but users discover that 20% of the expected records are missing. How would you investigate and prevent this class of issue?

Q113. How would you implement automated data-quality gates so that bad data does not reach curated datasets?

Q114. How would you design a framework for tracking data-quality rules and their results across hundreds of datasets?

Q115. How would you handle a source schema change that silently converts a numeric column into text?

Q116. How would you detect unexpected null spikes or distribution changes in a production dataset?

Q117. How would you design alerting for data freshness so business users are informed before reports become stale?

Q118. How would you quarantine invalid records while allowing valid records to continue through the pipeline?

Q119. How would you implement end-to-end lineage from source systems through transformation layers to BI reports?

Q120. How would you establish data ownership and stewardship for datasets used by multiple business teams?

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Q121. How would you design security for sensitive customer, financial, or employee data across ADLS, ADF, Synapse, Databricks, and BI tools?

Q122. How would you use Microsoft Entra ID identities and managed identities to reduce the use of long-lived credentials in data pipelines?

Q123. A developer requests Owner access to a production data resource to troubleshoot a pipeline. How would you handle the request securely?

Q124. How would you design network isolation for an enterprise Azure data platform using VNets, private endpoints, and controlled egress?

Q125. A data service is reachable publicly even though the organization requires private connectivity. How would you identify and remediate the exposure?

Q126. How would you manage secrets and connection information using Azure Key Vault across multiple environments?

Q127. How would you design role-based access so engineers can operate pipelines without being able to read sensitive business data?

Q128. How would you investigate an unexpected access to a sensitive data lake path?

Q129. How would you implement encryption, auditing, and retention controls for regulated data?

Q130. How would you design separation of duties between data engineers, platform administrators, analysts, and security teams?

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Q131. How would you implement data discovery, classification, lineage, and governance using Microsoft Purview?

Q132. Your organization has thousands of datasets and nobody knows which ones are trusted. How would you establish a governed data catalog?

Q133. How would you use Purview to identify sensitive information and improve governance without blocking legitimate analytics?

Q134. How would you design ownership, glossary, certification, and lifecycle processes for enterprise datasets?

Q135. A business user finds two datasets with different definitions of 'active customer.' How would you resolve the issue through data governance?

Q136. How would you design a governance model that supports self-service analytics while maintaining security and compliance?

Q137. How would you monitor and audit data access across multiple Azure subscriptions and data services?

Q138. How would you design policies that prevent teams from creating unmanaged or insecure data resources?

Q139. How would you establish standards for naming, tagging, retention, ownership, and access across an Azure data estate?

Q140. How would you measure the maturity and effectiveness of an enterprise data governance program?

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Q141. How would you implement CI/CD for ADF, Synapse, Databricks notebooks, and other data engineering assets across development, test, and production?

Q142. A production deployment introduced an incorrect transformation and business reports are wrong. How would you contain the issue, restore trusted data, and prevent recurrence?

Q143. How would you design infrastructure as code for an Azure data platform using Terraform or Bicep?

Q144. A deployment works in development but fails in production because of configuration differences. How would you redesign environment configuration management?

Q145. How would you design an end-to-end orchestration framework that coordinates ingestion, transformation, validation, and publishing across multiple Azure services?

Q146. A critical pipeline must process data within a strict SLA. How would you design monitoring, alerting, retry, escalation, and recovery procedures around that SLA?

Q147. How would you migrate a large legacy ETL estate to Azure while maintaining business continuity and validating that the new platform produces equivalent results?

Q148. How would you investigate a sudden increase in Azure data-platform costs and identify which workloads or design choices are responsible?

Q149. How would you design a platform that supports both batch and streaming analytics while keeping data contracts, governance, security, and operational ownership clear?

Q150. Walk through how you would investigate a critical production data incident—from detection and containment through data recovery, root-cause analysis, validation, and prevention of recurrence.